

NGUYEN KHANH HOA

QA ENGINEER (MANUAL & AUTOMATION)

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Location: Tan Binh District, HCMC, Vietnam

PROFESSIONAL SUMMARY

Dedicated QA Engineer with **2+ years of hands-on experience** in Software Quality Assurance, specializing in both Manual and Automation Testing for complex web applications and e-commerce platforms, complemented by a solid frontend development foundation.

Proven track record in designing test strategies, establishing automation frameworks from scratch, and integrating testing into CI/CD pipelines within Agile Scrum environments. Highly proficient in **Java** (Selenium WebDriver, TestNG, POM) and experienced in utilizing modern web technologies (**JavaScript/TypeScript, ReactJS**) to implement Shift-Left testing strategies. Possess hands-on capabilities in API testing, Performance testing (JMeter), and modern automation tools like **Playwright** and **Python**.

Adept at bridging the gap between development and QA teams to ensure high-quality software delivery, scalability, and optimal user experiences.

TECHNICAL SKILLS & CORE COMPETENCIES

Testing Methodologies	Manual & Automated Testing, Shift-Left Testing, Regression, Exploratory, Smoke/Sanity Testing.
Test Engineering	Test Strategy & Planning, Framework Design (Data-Driven, POM), API Verification, Performance/Load Testing.
Process & Delivery	Agile/Scrum, SDLC/STLC, CI/CD Integration, Defect Lifecycle Management, Root Cause Analysis.
Programming & Scripting	Proficient: Java (OOP, Core Java for Test Automation), JavaScript / TypeScript, HTML/CSS. Familiar/Applied: Python (Scripting & Data processing), SQL (Data validation).
Test Automation & Tools	UI Automation: Selenium WebDriver, Playwright (TypeScript/JS approach). Frameworks & Design: TestNG, Page Object Model (POM), Data-Driven Testing. API Testing: Postman (Collections, Environment variables, Automated assertions). Performance Testing: Apache JMeter (Thread Groups, Assertions, Throughput & Bottleneck Analysis).
AI-Assisted Testing	Skilled in using AI tools to generate, optimize, and maintain test cases and improve testing efficiency.
DevOps, CI/CD & Environment	Version Control: Git, GitHub. CI/CD: GitHub Actions (Automated test execution on Pull Requests). Project Management: Jira, Confluence. Databases: MySQL / PostgreSQL / SQL Server.

WORK EXPERIENCE

AGEST VIETNAM

06/2024 – Present

QA Engineer (Manual & Automation)

2 years+

- Led the end-to-end testing lifecycle for enterprise web applications, collaborating closely with POs and Devs to refine acceptance criteria and prevent defects in the early stages (Shift-Left).
- Architected and maintained a robust scalable UI automation framework using **Java, Selenium WebDriver, and TestNG** applying the Page Object Model (POM).
- Integrated automated regression test suites into **GitHub Actions**, reducing manual regression time by **65%** and enabling continuous feedback for the development team.
- Authored and maintained API test scripts using Postman to validate business logic independently from the UI.
- Conducted Performance & Load testing utilizing **Apache JMeter**, successfully identifying memory leaks and DB query bottlenecks under concurrent user loads of 500+.
- Mentored junior testers/interns on Agile processes, test case optimization, and basic Java automation.

- Developed responsive UI components using **ReactJS, HTML5, and CSS3**, gaining first-hand understanding of frontend architecture and common UI defect patterns.
- Actively participated in debugging and testing phases, identifying and resolving cross-browser compatibility issues and reducing UI-related bugs by **20%** before QA handover.
- Built a deep working knowledge of **DOM structure, RESTful APIs, and browser developer tools** (Network, Console, Elements) — directly applicable to UI automation and defect root cause analysis.
- Collaborated with backend developers on API integration, strengthening understanding of request/response workflows critical for API testing.

FEATURED PROJECTS

Safe Railway – Enterprise Ticketing & Logistics System

06/2024 – 02/2025

Role: Automation & Performance QA Engineer

Tech Stack: Java, Selenium, TestNG, JMeter, GitHub Actions, MySQL.

Description: A high-traffic web-based railway management and booking system serving **10,000+ daily active users**, featuring dynamic pricing, real-time seat mapping, and integrated payment gateways.

- Designed a comprehensive test strategy covering functional, integration, and performance layers.
- Developed an automated regression suite from scratch (Selenium + Java) covering 100+ critical user paths (Search, Reserve, Payment Mock).
- Configured **GitHub Actions** to trigger automated tests upon every deployment to the staging environment.
- Configured JMeter load tests to simulate holiday-season traffic spikes (1000+ requests/sec), collaborating with backend devs to optimize the booking API response time from 3.5s to under 1.2s.

Omnichannel E-Commerce Platform

03/2025 – Present

Role: QA Engineer

Tech Stack: Playwright (TypeScript), Postman, Python (Data Mocking), Jira.

Description: A scalable omnichannel e-commerce platform supporting **5,000+ SKUs across 3 sales channels** (Web, Mobile, POS), integrating real-time inventory management, cart workflows, and multiple third-party logistics APIs.

- Transitioned the team to modern testing tools by piloting UI automation using **Playwright with TypeScript**, significantly improving test execution speed compared to legacy scripts.
- Conducted rigorous API testing via Postman to validate inventory synchronization across web and mobile platforms.
- Utilized Python scripts to automatically generate large volumes of mock test data for edge-case scenarios (e.g., flash sales, negative inventory logic).
- Performed extensive exploratory and boundary value testing to ensure a flawless checkout experience, resulting in a 99.5% bug-free release rate in production.

EDUCATION & CERTIFICATIONS

TON DUC THANG UNIVERSITY – Bachelor of Software Engineering | GPA: 7.61 / 10

- **Certification:** Aptis ESOL B1 (03/2024)
- **Certification:** Agile Development & Scrum Framework Professional (10/2023)